

THE WORLD'S FIRST COMPUTERIZED SCALE

ACW-M-1

1971年の夏、当社は高知県の農協様からピーマンの自動計量機の開発の依頼を受け、開発陣は試行錯誤の末、コンピュータによる「組合せ計量方式」を誕生させました。これは、8個の計量器で量られたピーマンの重さを中央演算装置で集計し、その中から合計150gになる組合せ(組合せパターン)は255通り、計量器が10個の場合は1023通り(10進法に選り出すという画期的なものでした)。

1972年12月に記念すべきコンピュータの第一号試作機が完成し、翌年1973年には開発依頼を受けていた高知県芸西村農協・丸二園芸様へのテスト導入が実現しました。以来、ウイスキー、スナックなどの食品業界をはじめ工業用その他で、多岐にわたって活躍することになりました。また海外からの引き合いも多く、「コンピュータスケール」は自動計量機の代名詞的存在となり、当社躍進の原動力となりました。本展示機は、広島県世羅町農協様からお譲りいただきました。

In the summer of 1971, Ishida received an inquiry for a scale which could weigh bell peppers of various sizes automatically. The Ishida Engineering team responded by inventing a computer-controlled weigher, which operated on the new principle of "Combination Calculation. The weigher worked like this: Bell peppers were weighed in eight separate hoopers (weigh head) and the data fed into a built-in calculation unit. The scale's computer then instantaneously selected the group of peppers (out of 255 possible combinations) closest to 150 grams. This system could accurately weigh vegetables at unhead of speeds and was the beginning of a true revolution in weighing technology.

In December of 1972, the world's first "multihead weigher" was ready to go to work and was installed at Maruni Engei in the Kochi Prefecture Geisel Village Agricultural Cooperative (which had made the original inquiry), for a trial run early the following year. This test was so successful that Ishida was soon filling orders for multihead units to process a growing variety of products such as sausages and snack foods. Since then, a wide range of weighers from Ishida, descended from the industry in countries on every continent.

After almost 25 years of continuous service, this machine was provided for exhibit by the Hiroshima Sera-cho Agricultural Cooperative. The weigher was still running perfectly at the time it was decommissioned.



Safety and Precautions

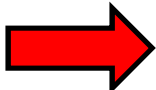
-  Never touch any electrical switches or buttons with wet or damp hands. Electrical shock may occur when the equipment is not properly grounded or when there is electrical leakage.
-  Do not operate the equipment with the cover of each part removed. Before starting operation of the equipment, make sure that all covers are securely shut and fastened. Do not touch any moving parts during operation. Failure to observe this warning may cause personal injury.
-  Personnel with long hair using this equipment should tie up their hair securely and all personnel must wear a cap or hat as well as clothes and shoes suitable for the production environment.
Unbound long hair or inappropriate clothing may become caught in moving parts, and injuries may result.
-  Before performing any maintenance or inspection work on the equipment, especially work which is not specifically indicated in this manual, shut off and lock the main power switch.
The person performing the maintenance should keep the key in his possession while performing the work.
Injury or electrical shock may result if the equipment is turned on by other person while maintenance, etc. work is being performed.
-  Turn off the main power switch then wait at least 3 minutes before performing any maintenance or inspection work on the equipment.
Residual charge may remain in the machine even after the power has been turned off. Also, turn off the power of any upstream/downstream units which are connected or adjacent to the equipment, otherwise there is a danger of electrical shock.

Safety and Precautions

WARNING

-  Before cleaning the equipment, shut off and lock the main power switch. The person performing the cleaning should keep the key in his/her possession while cleaning. Electrical shock or injury may result if the equipment is turned on by other person.

CAUTION

-  Do not use the same power supply for devices which may emit noise. Doing so may result in malfunction or damage.
-  Do not apply insecticide or other foreign substances inside the main body, terminal boxes, motor boxes or other enclosed equipment compartments. Doing so may cause machine malfunction or damage.
-   Do not apply excessive load to the weigh hoppers. Doing so may damage the weigh mechanism.
-  Do not install wiring which permits load-related power fluctuations of more than $\pm 10\%$. Overload may result in malfunction or damage.
-  Weighing fresh meat or cut vegetables with this equipment under improper conditions may result in hygienic hazards. Perform cleaning under the proper procedures according to product conditions and weighing methods. Read through and understand Chapter 9 before performing cleaning.

Safety and Precautions

-  If maintenance or inspection work is to be performed with the main power switch ON, clearly indicate this situation by posting a sign in the work area. This is to prevent other personnel from accidentally starting up the equipment.

CAUTION

-  Do not press the operation panel of the remote control unit using a ball-point pen or other pointed object.
Using a pointed object can damage the operation panel. Use finger to press the operation panel of the remote control unit.
-  When performing maintenance and inspection work on the upper part of the equipment, use a sturdy ladder or foothold to avoid a hazardous fall.
-  When starting production, make sure that there is no object such as a tool on the machine.
If objects fall into moving parts, damage to the equipment may result.

Principles of Combination Weighing

A. Dispersion Table (Feeder)

Product is fed from a crossfeeder onto the dispersion table. The amount of product fed to the dispersion table is regulated by a load cell located under the dispersion feeder assembly. The dispersion table is vibrated by the dispersion feeder amplitude and time to move the product onto the radial pans.

B. Radial Troughs (Feeder)

Precisely tuned, vibrating radial feeders distribute product from the radial pans to the pool hoppers. The radial feeders are controlled by vibration amplitude and time.

C. Pool Hoppers

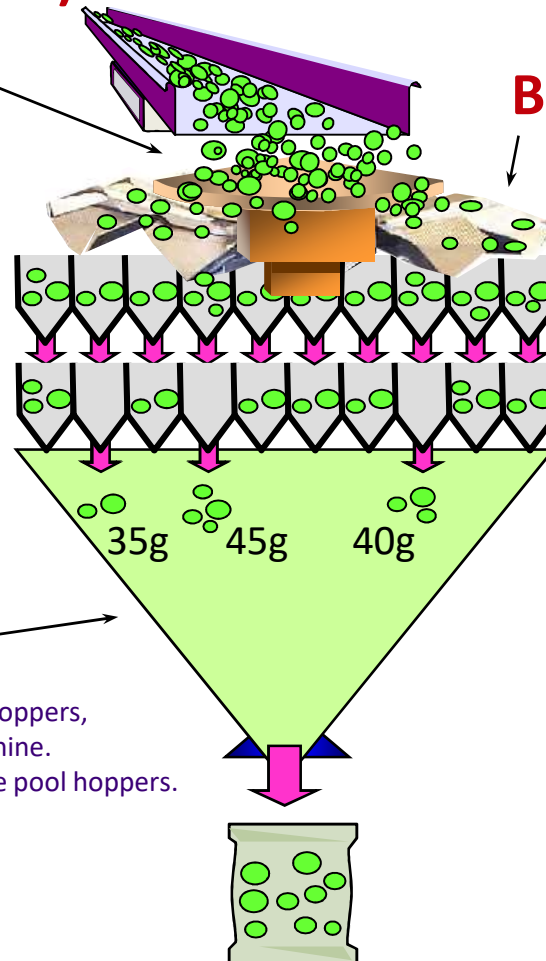
Product fed to the appropriate pool hoppers, is held before it is released to the respective weigh hoppers. This provides a transition between the radial feeders and the weigh hoppers.

D. Weigh Hoppers

Load cells weigh the product in each weigh hopper. A microprocessor scans the available weights and selects the hoppers whose combined weights equal or come closest to the required package weight.

E. Discharge Chutes

Product released from the weigh hoppers and booster hoppers, falls through the discharge chutes to the packaging machine. Emptied weigh hoppers are refilled from their respective pool hoppers.



2^x-1

$$2^{14} - 1 = 16,383$$

$$2^{13} - 1 = 8,191$$

$$2^{12} - 1 = 4,095$$

$$2^{11} - 1 = 2,047$$

$$2^{10} - 1 = 1,023$$

$$2^9 - 1 = 511$$

$$2^8 - 1 = 255$$

$$2^7 - 1 = 127$$

$$2^6 - 1 = 63$$

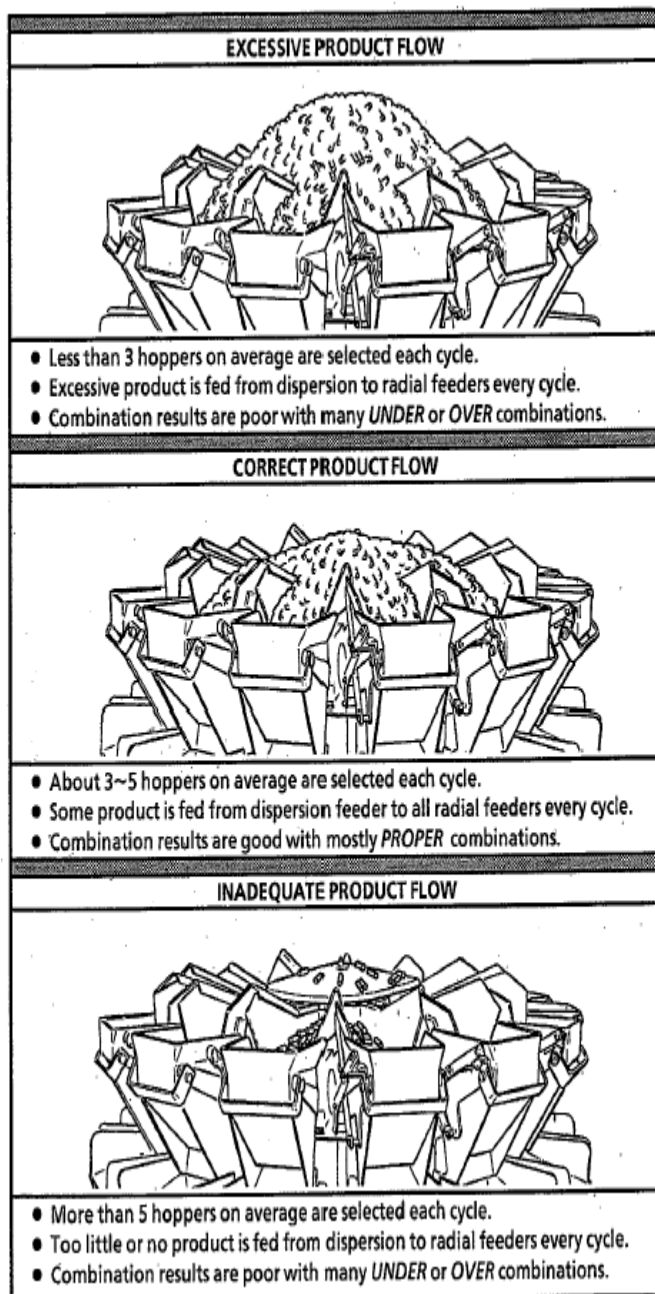
$$2^5 - 1 = 31$$

$$2^4 - 1 = 15$$

$$2^3 - 1 = 7$$

$$2^2 - 1 = 3$$

$$2^1 - 1 = 1$$



Overscale:

1. *The weight in a single Weigh Hopper exceeds that of the Upper Limit weight;
For example: Target Weight is 10grams and the Upper Limit weight is 1gram, however, the weight in a weigh hopper is 11.3grams.
2. ***AND/OR**, The weight in a single Weigh Hopper that exceeds the range of the Load Cell capacity;
For example: The load cell range is set to 400grams, however, the weight in the weigh hopper is 406grams.
3. ***AND/OR**, When performing a multiple dump, the weight in a single Weigh Hopper exceeds the Upper Limit weight when dividing the weight into the amount of "dumps" needed to perform a combination calculation.
For example: The dump count is set to two dumps. The target weight is 10grams and the Upper Limit weight is 1gram. The weight in a weigh hopper is 6.2grams.

Overweight:

*The weight in a combination of Weigh Hoppers exceeds that of the Target Weight plus the Upper Limit. As such; the weigher is not able to perform a Proper Combination Calculation as the closest combination exceeds the Upper Limit Target Weight. The default setting is set to four Overweight's in a consecutive manner. If during any of the four overweight calculation a proper or under weight is found, then the overweight buffer is cleared and new calculations are resumed.